RF Argon & Oxygen Plasma Surface Activation Chamber

Authoritative Model Technical Manual & Diagnostic Reference | VAI-MAN-PC-001

NOTICE: SYNTHETIC PROTOTYPE TECHNICAL MANUAL

This document is artificially generated for the VectorAI demonstration and software development.

The specifications, thresholds, service-life values, maintenance intervals, and operating parameters are synthetic and must not be used for real industrial equipment operation or maintenance.

1. Machine Overview & Manufacturing Context

Equipment Model:	RF Argon & Oxygen Plasma Surface Activation Chamber	**Document ID:**	VAI-MAN-PC-001
Machine Type Key:	plasma-cleaner	**Cleanroom Bay / Stage:**	Bay 3A: Plasma Activation
Document Version:	1.0	**Release Date:**	2026-08-27

Low-pressure 13.56 MHz RF plasma treatment chamber for micro-cleaning organic contaminants and activating metal bond pads (Au, Cu, Al) on die-attached leadframes prior to wire bonding.

Process Flow: Magazines of leadframes are loaded into an automated vacuum chamber. A dry mechanical pump evicts air down to < 10 Pa. Pure argon and oxygen gases are energized by a 13.56 MHz RF match network into a glow-discharge plasma, stripping organic monolayers and increasing copper surface energy to > 72 dyn/cm.

2. Major Subsystems & Critical Components

Subsystem / Component	Primary Function	Key Parameters	Degradation Indicators
RF Auto-Match Network	Matches generator 50Ω impedance to dynamic plasma chamber impedance.	Reflected power (W), forward power (W), tune/load capacitor pos (%).	Reflected power surge (>30W), tuning hunt time (>5s).
Dry Vacuum Pump Stack	Evacuates chamber down to operating base pressure (10-100 Pa).	Chamber base pressure (Pa), pump vibration (mm/s), pump temp (°C).	Pump vibration rise (>0.4 mm/s), base pressure degradation.
Process Gas MFCs	Regulates precise mass flow of ultra-pure Argon (Ar) and Oxygen (O_2).	Flow rate (sccm), gas inlet pressure (bar).	MFC flow hunt, solenoid response drift.
Chamber Door Fluoroelastomer Seal	Maintains hermetic vacuum seal during high-frequency plasma cycle.	Chamber leak-up rate (Pa/min), door clamp pressure.	Leak rate > 2 Pa/min, visual elastomer cracking.

3. Sensor Telemetry & Threshold Matrix

Threshold boundaries configured for real-time telemetry monitoring and automatic anomaly triggers:

Sensor Name & ID	Unit	Normal Range	Warning Range	Critical Range	Direction
RF Reflected Power rf_reflected_power	W	5 – 22	**22 – 38**	**38 – 100**	Higher is Worse
Chamber Pressure chamber_pressure	kPa	70 – 95	**95 – 120**	**120 – 150**	Higher is Worse

4. Deterministic RUL Model (Weighted Degradation)

Base Useful Life: 2,800 operating hours. **Model:** RUL = BaseLife * (1 - (0.50*wear_rf_refl + 0.30*wear_press + 0.20*wear_pump_vib))

Parameter	Sensor ID	Unit	Weight	Healthy Limit	Critical Limit	Direction
RF Reflected Power	rf_reflected_power	W	**0.50**	18	45	HIGHER IS WORSE
Chamber Pressure	chamber_pressure	kPa	**0.30**	85	125	HIGHER IS WORSE
Pump Vibration	vibration_vacuum_pump	mm/s	**0.20**	0.2	0.8	HIGHER IS WORSE
TOTAL WEIGHT SUM	-	-	**1.00 (100%)**	-	-	VERIFIED OK

5. Troubleshooting & Anomaly Symptoms Matrix

Symptom & ID	Severity	Related Sensors	Possible Root Causes	Recommended Corrective Action
Low Surface Energy (< 52 dyn/cm on Copper) SYM-PC-001	**HIGH**	rf_reflected_power, chamber_pressure	• RF match network out of tune • Chamber O-ring leak • O2 gas MFC blocked	Tune RF matching network and measure contact angle with goniometer.

6. Authoritative Diagnostic Knowledge Base (Failure Scenarios)

Scenario 1: RF Auto-Match Capacitor Drift & Vacuum O-Ring Degradation (SCEN-PC-001)	Severity: WARNING
Telemetry Sensor Pattern:	rf_reflected_power > 35W, chamber_pressure > 105 kPa
Possible Root Causes:	Vacuum variable capacitor electrode oxidation | Chamber seal micro-permeation
Recommended Corrective Action:	Re-tune RF matching network capacitor and replace fluoroelastomer chamber seal.
Verification & Recovery Steps:	Execute helium leak detector test (< 1x10^-8 mbar*L/s) > Run water droplet contact angle test (< 15 degrees)